Тема 1.1 Проверка системных требований перед установкой программного обеспечения

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1. Цель работы

Чтобы определить, подходит ли компьютер для установки выбранной программы, необходимо сравнить системные требования с характеристиками ПК.

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2. Выбранная программа

Название программы: Visual Studio Code
Источник: официальный сайт Microsoft

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3. Системные требования программы

Минимальные требования:

- **Процессор:** 1.6 GHz или выше
- **Оперативная память:** 1 GB
- **Свободное место на диске:** 200 MB
- **Операционная система:** Windows 10 или новее
- **Интернет:** требуется для загрузки и обновлений

Рекомендуемые требования:

- **Процессор:** 2 GHz и выше
- **Оперативная память:** 4 GB и более
- **Свободное место на диске:** 1 GB
- **Операционная система:** Windows 11

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4. Характеристики моего ноутбука

- **Процессор:** Intel Core i5 2.5 GHz
- **Оперативная память:** 8 GB
- **Свободное место на диске (C:):** 100 GB
- **Операционная система:** Windows 10 Pro

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5. Сравнение требований и характеристик

Параметр	Требуется	Есть на ПК	Соответствие
Процессор	1.6 GHz	2.5 GHz	Да
RAM	1 GB	8 GB	Да
Диск	200 MB	100 GB	Да
ОС	Windows 10	Windows 10 Pro	Да

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6. Вывод

Мой компьютер полностью соответствует минимальным и даже рекомендуемым требованиям программы Visual Studio Code. Все основные параметры (процессор, оперативная память и свободное место) превышают необходимые значения. Это значит, что программу можно установить без проблем, и она будет работать стабильно и быстро.

Topic 1.2 Installation of Application Software

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1. What Is Application Software?

Application software refers to programs designed to help users perform specific tasks.
Examples include web browsers, text editors, media players, and graphic design programs.
In this work, I installed a web browser.

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2. Selected Application

Application name: Google Chrome
Installation type: From installation file
Source: Official website

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3. Preparation Before Installation

Before installation, I:

1. Checked system requirements
2. Checked free disk space
3. Closed unnecessary programs

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4. Installation Process

Step 1: I downloaded the installer file.
Step 2: I double-clicked the file to start installation.
Step 3: The installation started automatically.
Step 4: The program downloaded necessary files from the Internet.
Step 5: After installation, the browser opened automatically.

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5. Installation Folder

The program was installed in the default folder:

C:\Program Files\Google\Chrome\

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6. Problems During Installation

There were no problems.
The installation was completed successfully.

Possible problems could be:

1. Not enough disk space
2. No Internet connection
3. Old operating system

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7. Conclusion

Application software is used for everyday tasks.
The installation process is simple and clear.
I successfully installed Google Chrome, and it works correctly on my computer.

Topic 1.3 Installation of System Software

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1. Operating System Name and Version

Operating System: Windows 10
Version: Windows 10 Pro (64-bit)

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2. System Requirements

Minimum requirements:

- Processor: 1 GHz or faster
- RAM: 2 GB
- Disk space: 20 GB
- BIOS or UEFI firmware

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3. Installation Method

The operating system was installed in a virtual machine using
Oracle VM VirtualBox.

I created a new virtual machine, selected Windows 10 (64-bit),
allocated 4 GB RAM and 50 GB disk space, and attached the ISO file.

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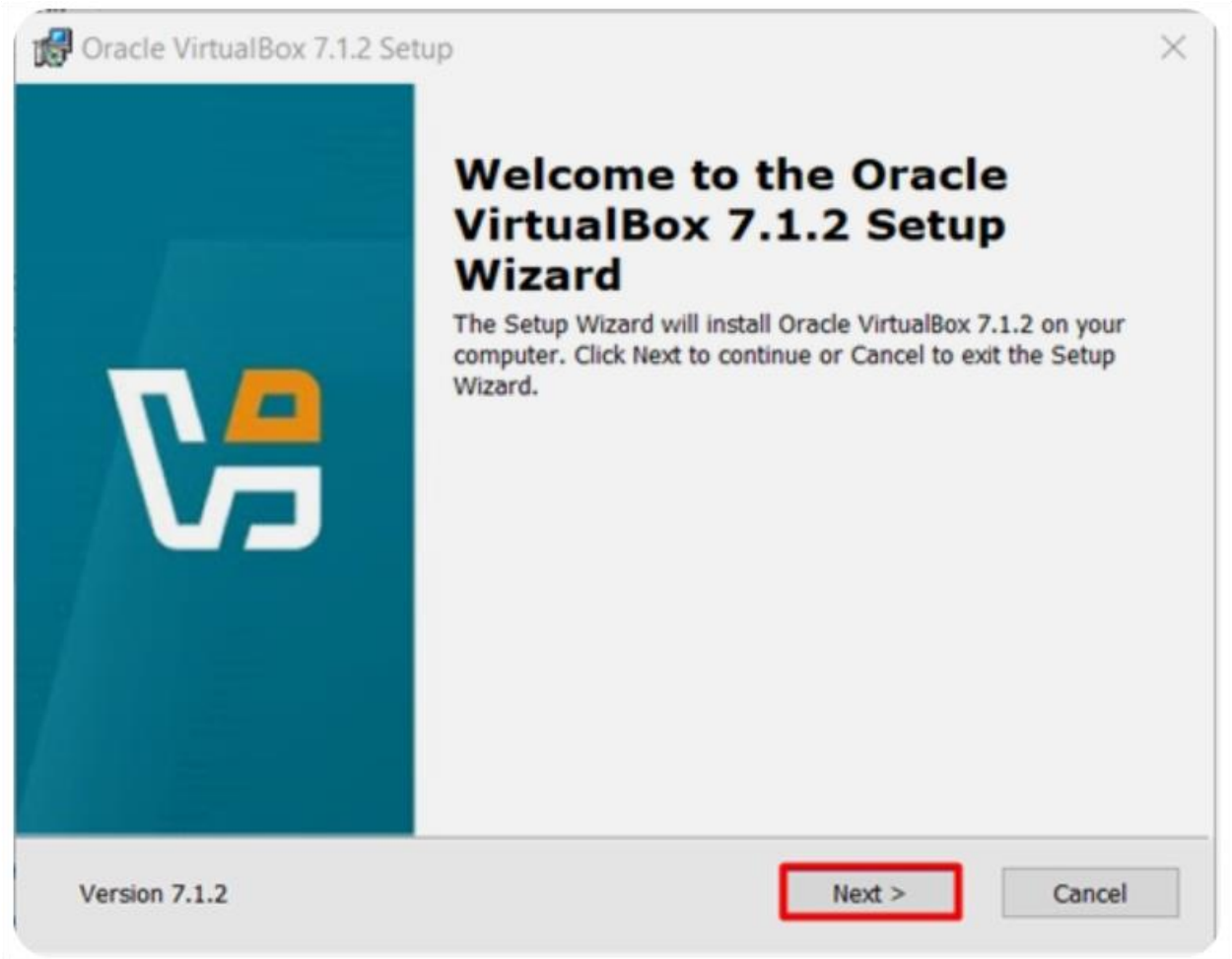
4. Installation Steps

1 Boot Selection

The virtual machine started from the ISO file.
The Windows installation screen appeared.



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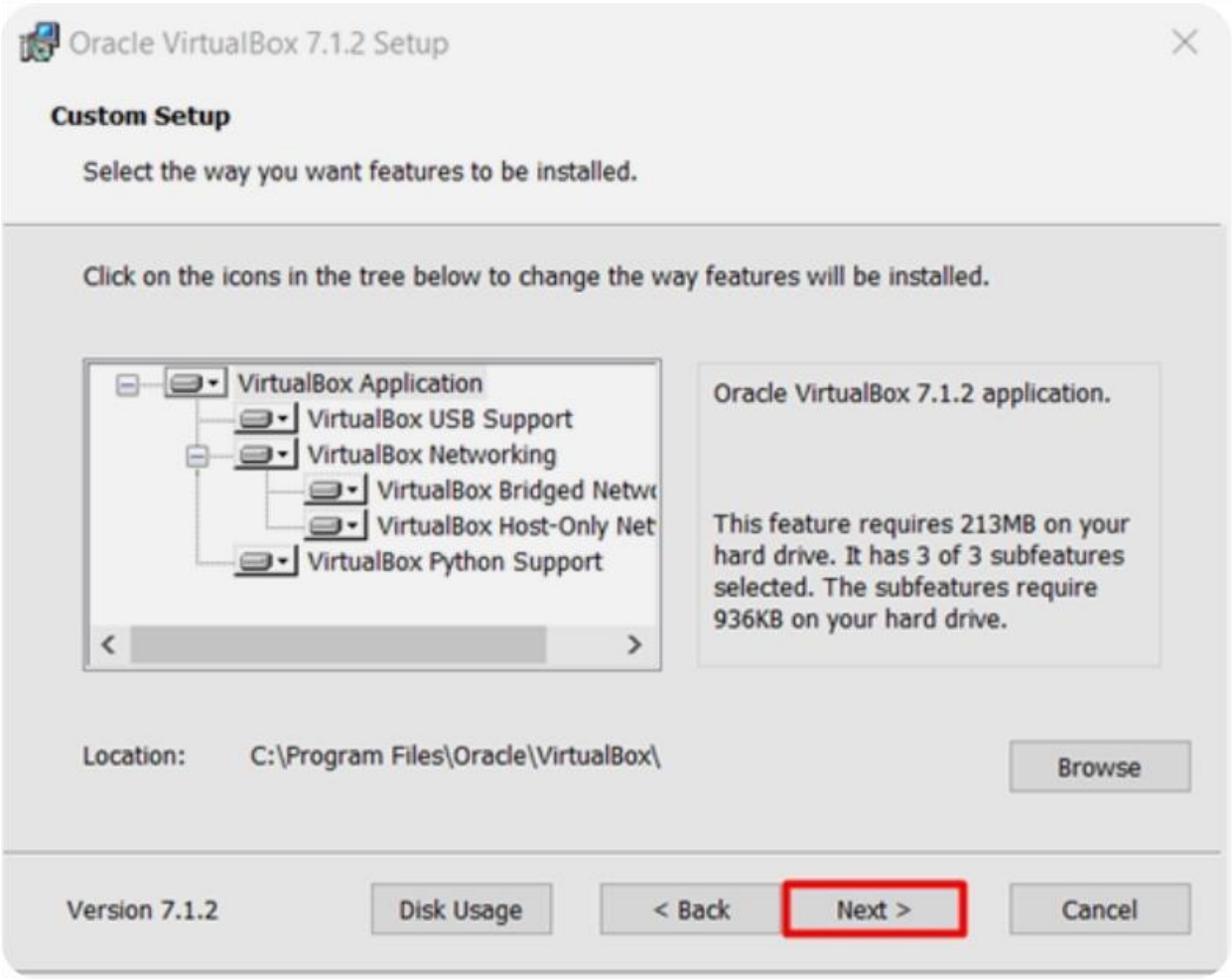
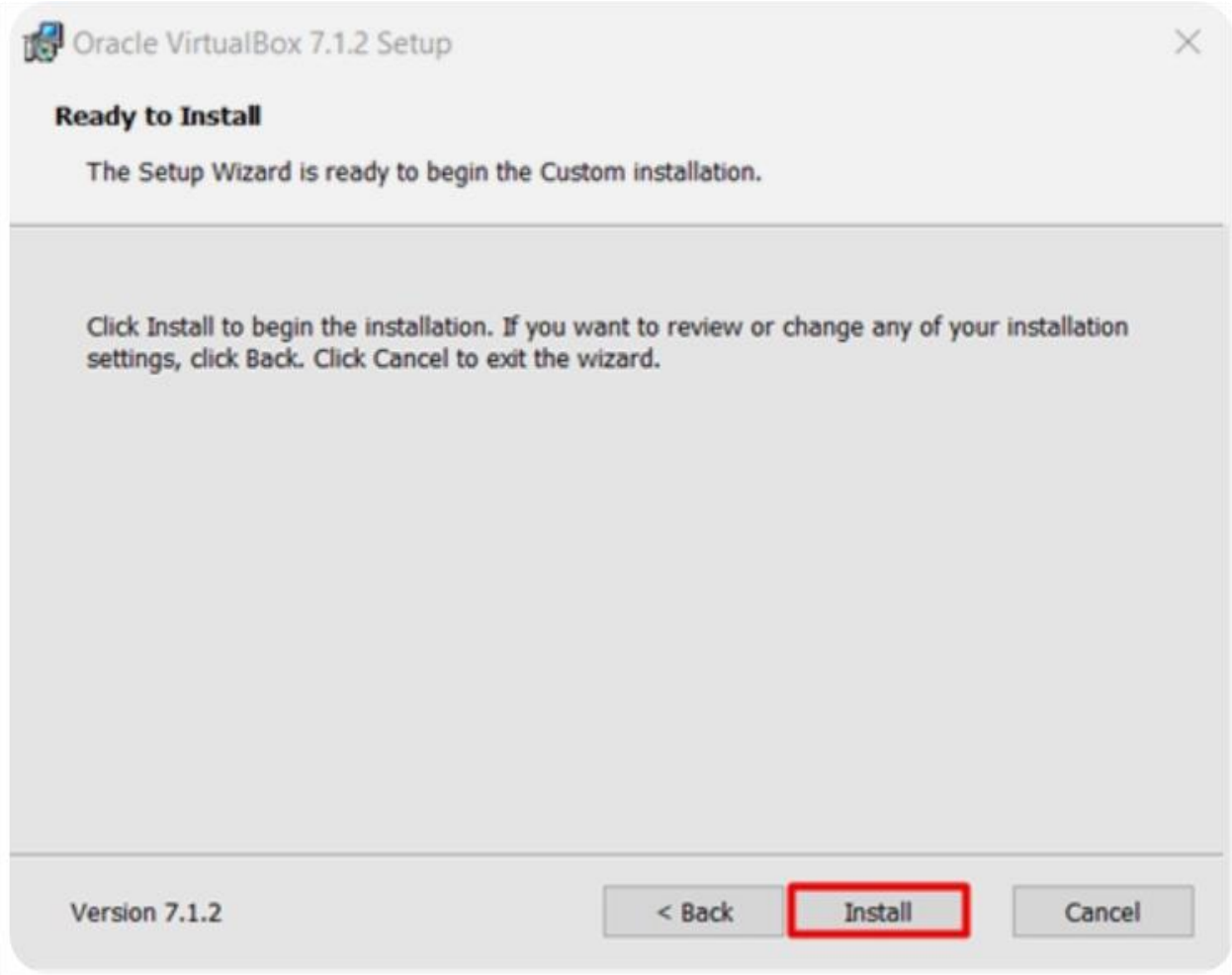
2 Disk Partitioning

I selected the virtual disk and formatted it.
Then I chose it for installation.



3 Installation Process

Windows copied files and installed system components.
The progress bar showed the installation status.



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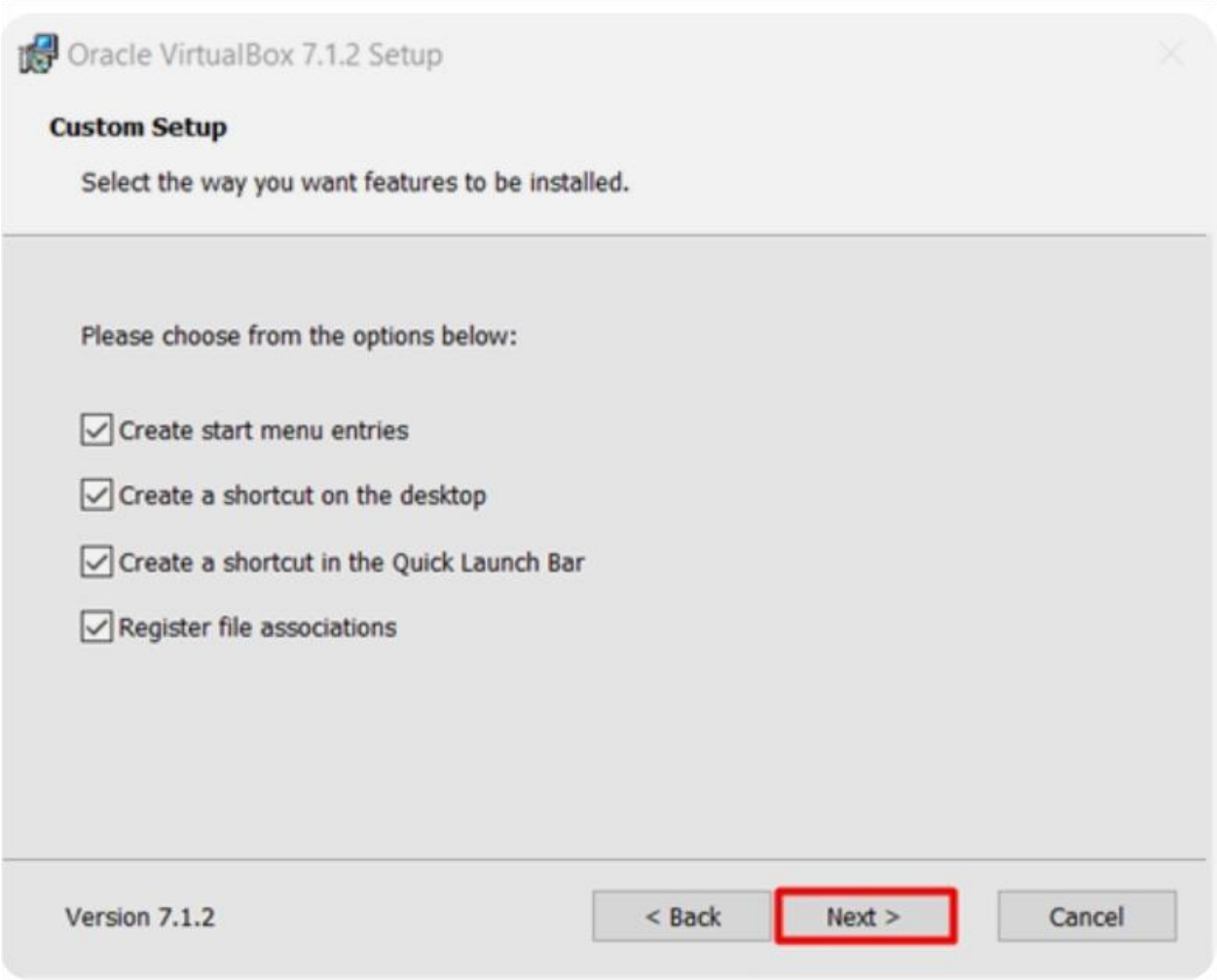
4 Initial Setup

Then I:

- Selected language and region
- Created a user account
- Set a password
- Connected to the network



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5. Post-Installation Configuration

After installation, I:

1. Checked system updates
2. Checked drivers
3. Configured basic settings

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6. Conclusion

System software controls hardware and allows the computer to work properly.
Installing an operating system includes several important steps such as booting, disk selection, file copying, and initial setup.
The installation was successful, and the system works correctly.

Topic 1.4 Installation of Device Drivers

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1. What a Device Driver Is

A driver is a program that helps the computer communicate with hardware.
Without drivers, devices like printers, video cards, or network adapters won't work.

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2. Types of Drivers

- **Built-in** – installed automatically by the system.
- **Manufacturer** – from the device maker for full features.
- **Generic** – basic functions, limited.
- **Signed** – verified by the system for security.

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3. When to Install Drivers

- After installing a new OS
- When connecting new hardware
- If a device is not recognized or works wrong
- To fix performance problems

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4. How to Install Drivers

1. **Automatic** – system installs itself.
2. **Device Manager** – select device - Update driver - search automatically or manually.
3. **From manufacturer** – download and run installer - restart.
4. **From USB or disk** – run installer from media.

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5. Manual Installation Steps

1. Find hardware model
2. Go to official website
3. Download correct driver for your OS (32/64-bit)
4. Install driver or use Device Manager
5. Restart computer

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6. Problems and Fixes

- Wrong version - reinstall correct one
- OS incompatible - find compatible driver
- Corrupted files - download again

- Conflicts - rollback or reinstall

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7. Safety Rules

- Only download from official sites
- Avoid unknown driver packs
- Make a restore point before installing
- Check OS compatibility
- Restart after installation

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8. Conclusion

Drivers are essential for hardware to function properly.
They can be installed either automatically or manually.
Following proper guidelines and troubleshooting steps helps resolve issues and ensures devices operate smoothly.

Topic 1.5 Configuration of Installed Software Parameters

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1. Introduction

After installing software, it needs to be configured to suit user needs.
Proper configuration makes software work better, faster, and safer.

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2. Types of Software Settings

- **General settings:** language, theme, notifications
- **Performance settings:** memory usage, hardware acceleration
- **Security settings:** passwords, firewall
- **Network settings:** proxy, connection options
- **Update settings:** automatic or manual updates

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3. Why Configuration Is Important

Proper configuration:

- Improves performance
- Enhances security
- Uses resources efficiently

- Reduces conflicts
- Adapts software to user needs

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4. Methods of Configuration

- Using built-in settings menu
- Editing configuration files
- Using Control Panel or system tools
- Using administrative tools

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5. Example Configuration Steps

Web browser example:

- Set default search engine
- Configure homepage
- Adjust privacy settings
- Enable automatic updates
- Manage extensions

Antivirus example:

- Enable real-time protection
- Schedule automatic scans
- Configure firewall rules
- Update virus database

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6. Common Configuration Errors

- Turning off important security features
- Wrong network settings
- Using too many resources
- Incorrect permission settings

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7. Conclusion

Software configuration is essential for ensuring that programs function correctly and securely.
By carefully adjusting settings, users can enhance performance, strengthen security, and improve overall usability.
Understanding common mistakes allows users to prevent issues and maintain reliable software operation.